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Mini Project

Synopsis

ON

"AI-Powered Personalized Digital Twin for Cardiovascular Treatment Simulation"

submitted in the partial fulfilment of the requirements for V semester, Bachelor of Engineering in Computer Science and Engineering

By

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Table of Contents

Contents	Page no
Introduction	3
Literature Survey	5
Limitations of the Existing Works	10
Problem Statement	13
Objectives	14
Proposed Design (If any)	15
Expected Outcome	18
References	19

INTRODUCTION

Background and Domain Overview

Cardiovascular diseases (CVDs) remain the leading cause of global morbidity and mortality, placing an immense burden on modern healthcare systems. Despite substantial advances in pharmacological therapies, medical imaging, and diagnostic devices, standard clinical practice in cardiology still relies heavily on generalized, population-level evidence. Clinical decisions, dosage calibrations, and drug selections are typically guided by standardized protocols that treat diverse patient cohorts as homogeneous entities.

However, human biological systems exhibit extraordinary inter-individual heterogeneity. Variations in genetics, arterial elasticity, metabolic clearance rates, lifestyle factors, and comorbidities mean that two patients receiving identical diagnoses and treatments will often experience vastly different clinical trajectories. In fast-paced clinical settings, this biological variability frequently leads to a reactive, "trial-and-error" approach to prescription management. Adjusting antihypertensive or ionotropic regimens often requires weeks of observation, during which patients remain exposed to sub-optimal therapeutic outcomes or dangerous Adverse Drug Reactions (ADRs).

The Emerging Paradigm of Digital Twins in Medicine

To bridge the gap between population-based clinical guidelines and individual biological realities, computational medicine has introduced the concept of the **Medical Digital Twin**. Originating in aerospace and advanced industrial engineering, a digital twin is a dynamic, multi-scale computational replica of a physical asset that continuously updates itself using real-time sensor data. Translated to healthcare, a Cardiac Digital Twin forms an evolving virtual avatar of a patient's cardiovascular system, integrating baseline Electronic Health Record (EHR) parameters, diagnostic imaging, and longitudinal vital signals.

While early medical digital twins relied on multi-scale 3D Computational Fluid Dynamics (CFD) and Finite Element Analysis (FEA), these models present severe operational bottlenecks. 3D fluid-structure interaction simulations require hours—or even days—of high-performance computing (HPC) execution time to model a single cardiac cycle. This high computational latency renders full 3D cardiac simulations completely unviable for real-time, point-of-care clinical decision support.

Project Motivation: The DigiTwin Approach

The **DigiTwin** project was conceived to overcome these computational and operational limitations by engineering a low-latency, hybrid physics-AI Cardiac Digital Twin platform. Driven by the core operational philosophy of **"Simulate before you prescribe,"** DigiTwin equips clinicians with a real-time *in silico* drug titration sandbox.

Instead of relying on heavy 3D spatial models, DigiTwin adopts a reduced-order **Lumped 0D Pharmacokinetic-Pharmacodynamic (PK-PD)** differential solver coupled with deep recurrent neural networks (1D CNN, GRU/LSTM) and gradient boosting classifiers (XGBoost). By substituting high-dimensional spatial calculations with 0D vascular

circuit modeling and AI temporal forecasting, DigiTwin reduces simulation latency from hours down to **under 2 seconds** on standard web infrastructure.

Furthermore, DigiTwin is explicitly designed to address regional demographic gaps. Standard clinical prediction tools are overwhelmingly trained on homogeneous Western datasets (such as MIMIC-IV), often failing to capture the unique genetic, ancestral, dietary, and lifestyle baselines of diverse populations. By incorporating localized parameter-seeding algorithms tailored specifically to South Asian/Indian clinical profiles, DigiTwin delivers explainable, demographics-aware precision cardiology.

LITERATURE SURVEY

1. Digital Twins in Healthcare: A Review of AI-Powered Practical Applications Across Health Domains

- Journal / Publisher: *Journal of Big Data* (Springer Open, 2025)
- Authors: Ziad Elgammal, M. Taleb Albrijawi, Reda Alhajj
- Why it is Important:
    - Establishes the Multi-Domain Paradigm: Proves that digital twins are transitioning from speculative engineering concepts into practical clinical tools across cardiology, oncology, and endocrinology.
    - Defines Continuous Data Fusion: Synthesizes how integrating Electronic Health Records (EHRs), medical imaging, and continuous IoT bio-sensors forms dynamic patient avatars that update over time rather than remaining static snapshots.
    - Project Relevance: Provides foundational justification for DigiTwin's multi-modal data architecture.

2. Digital Cardiovascular Twins, AI Agents, and Sensor Data: A Narrative Review from System Architecture to Proactive Heart Health

- Journal / Publisher: *Sensors* (MDPI, 2025)
- Authors: Nurdaulet Tasmurzayev et al.
- Why it is Important:
    - Shift to Proactive Cardiology: Demonstrates that streaming physiological telemetry (ECG, PPG, blood pressure) enables the early detection of subclinical cardiac pathologies long before tissue damage occurs.
    - Multi-Layer Architecture: Details the systemic framework connecting wearable sensors, analytics engines, and autonomous decision-support agents for continuous risk monitoring.
    - Project Relevance: Serves as the blueprint for DigiTwin's real-time telemetry ingestion and proactive alert infrastructure.

3. A Hybrid AI and Simulation Framework for Patient-Specific Cardiovascular Digital Twins

- Journal / Publisher: *IEEE Access* (IEEE, 2025)
- Authors: Harshit Sharma & Simran Kaur
- Why it is Important:

- Solves the Compute Latency Barrier: Proves that combining mechanistic differential equations (physics) with deep neural network surrogates (AI) drastically accelerates simulation execution speed without losing physiological accuracy.
- Real-Time Feasibility: Demonstrates that sub-second computational performance is achievable, bringing virtual patient modeling out of high-performance computing (HPC) labs and directly to clinical dashboards.
- Project Relevance: Directly validates DigiTwin's Hybrid 0D PK-PD + ML core engine.

4. Building Digital Twins for Personalized Cardiovascular Medicine: Engineering Principles and Clinical Applications

- Journal / Publisher: *Computers in Biology and Medicine* (Elsevier, 2025)
- Authors: Claudio Chiastra et al.
- Why it is Important:
  - Establishes Mathematical Trade-offs: Systematically compares complex 3D Computational Fluid Dynamics (CFD) / Finite Element Analysis (FEA) against reduced-order 0D Lumped Parameter models.
  - Justifies Reduced-Order Modeling: Explains why 3D models (which take hours to execute) are impractical at the point of care, confirming that 0D vascular circuit modeling is the optimal choice for interactive clinical drug titration.
  - Project Relevance: Provides the explicit bio-engineering rationale for DigiTwin choosing a 0D Lumped differential solver over computationally heavy 3D spatial models.

5. Mathematical and Epidemiological Frameworks for Digital Twin Applications in Precision Healthcare

- Journal / Publisher: *JMIR Medical Informatics* (JMIR Publications, 2025)
- Author: Aurélien Vallée
- Why it is Important:
  - Rigorous Mathematical Foundation: Formulates ordinary differential equation (ODE) frameworks for simulating individual physiological dynamics and population-level health trends.
  - Explainable Mechanics: Establishes that deterministic mathematical formulas are necessary to keep decision-support systems bound to real physical laws.
  - Project Relevance: Grounds DigiTwin's differential physics solver (SciPy ODEs) in formal systems biology theory.

6. Artificial Intelligence and Digital Twin Technologies in Cardiovascular Disease Management

- Journal / Publisher: *World Academy of Sciences Journal* (Spandidos Publications, 2025)
- Authors: Sharon John et al.
- Why it is Important:
    - Proves Clinical ADR Reduction: Demonstrates that running virtual *in silico* drug trials before writing real prescriptions significantly reduces trial-and-error adjustments and prevents Adverse Drug Reactions (ADRs).
    - Dose Calibration Efficiency: Proves that personalized hemodynamic modeling helps clinicians achieve target therapeutic levels faster than traditional guidelines.
    - Project Relevance: Quantifies the clinical impact behind DigiTwin's primary motto: "Simulate before you prescribe."

7. From Polygenic Risk to Digital Twins: Integrating Multi-Omics for Cardiac Risk Stratification

- Journal / Publisher: *Frontiers in Cardiovascular Medicine* (Frontiers Media, 2026)
- Authors: Ramy Antoun et al.
- Why it is Important:
    - Integrates Omics & Multi-Scale Data: Shows how incorporating Polygenic Risk Scores (PRS) and transcriptomic profiles refines individual cardiac risk assessment and drug response predictions.
    - Addresses Population Heterogeneity: Highlights how genetic variations alter baseline cardiovascular risk across different ethnic and geographic groups.
    - Project Relevance: Supports DigiTwin's emphasis on overcoming Western demographic bias by incorporating tailored baseline parameters.

8. Digital Twins in Medicine: Simulating Human Biology for Personalized Treatment

- Journal / Publisher: *SSRN Electronic Journal* (Preprint, Elsevier, 2026)
- Author: Obi Blessing Obi
- Why it is Important:
    - Bridging Industrial Concepts to Biology: Outlines the overall conceptual architecture of medical digital twins, emphasizing multi-scale biological modeling and real-time bidirectional feedback loops.
    - Economic & Systems Impact: Explains how virtual patient models reduce emergency readmissions, cut healthcare costs, and transform healthcare from reactive treatment to preventive care.

- Project Relevance: Supplies broader clinical workflow and economic justification for hospital-wide digital twin adoption.

9. Digital Twin Ecosystems for Personalized Oncology: From Virtual Patients to Real-Time Clinical Intelligence

- Journal / Publisher: *IJERAMS* (Vol. 1, Issue 5, 2025)
- Authors: Dr. Harleen Kaur, Dr. Sameer Vyas, Dr. Pritam Nair, Mrs. Kavya Thomas, Dr. Neel Shah
- Why it is Important:
  - Cross-Domain Validation: Demonstrates that multi-layered digital twin ecosystems successfully integrate pathology, imaging, and lab data to track complex disease progression (tumor growth, immune response).
  - Real-Time Ecosystem Synchronization: Illustrates how continuous synchronization between physical patients and virtual avatars allows clinical teams to recognize therapeutic resistance early.
  - Project Relevance: Validates that DigiTwin's architectural framework scales beyond cardiology into broader multi-organ digital twin ecosystems.

10. Artificial Intelligence and Advanced Digital Health for Hypertension: Evolving Tools for Precision Cardiovascular Care

- Journal / Publisher: *Medicina* (MDPI, 2025)
- Authors: Ioannis Skalidis et al.
- Why it is Important:
  - Overcomes Therapeutic Inertia: Evaluates AI-guided titration platforms (e.g., CURATE.AI) and remote blood pressure monitoring (RBPM), showing clinically significant reductions in systolic blood pressure (5.3–14.2 mmHg).
  - Identifies Practical Hardware Gaps: Highlights that consumer-grade cuffless blood pressure sensors currently lack clinical accuracy (_+30mmHg error), proving that hybrid algorithmic modeling with validated inputs remains superior.
  - Project Relevance: Directly justifies DigiTwin's sub-2-second drug titration engine aimed at eliminating clinician dosage inertia.

LIMITATIONS OF EXISTING WORKS

- **Elgammal et al. (2025) – *Journal of Big Data***
    - Black-Box AI Dependency: Deep learning architectures lack biological mechanics, reducing physician trust during critical bedside care.
    - Data Synchronization Latency: Aggregating multi-source streams (EHRs, imaging, wearables) creates execution bottlenecks that prevent real-time utility.
    - Privacy Governance: Handling continuous health streams exposes critical dynamic data privacy and HIPAA/GDPR compliance risks.
- **Tasmurzayev et al. (2025) – *Sensors***
    - Telemetry Vulnerability: Relies heavily on continuous IoT streaming, making system accuracy susceptible to packet loss and signal dropouts.
    - Infrastructure Expense: High cloud data-ingestion and streaming costs limit deployment in budget-constrained public health systems.
    - Artifact Interference: Motion artifacts in consumer wearables frequently trigger false-positive clinical alerts.
- **Sharma & Kaur (2025) – *IEEE Access***
    - Demographic Bias: Physics surrogate networks are trained predominantly on Western cohorts, degrading accuracy for South Asian populations.
    - Synthetic Overfitting: Surrogate models trained on artificial ODE datasets struggle to adapt to noisy, non-linear real-world clinical anomalies.
    - UI Integration Gap: Focuses strictly on computational execution speed without offering an interactive web interface for clinical dosage calibration.
- **Chiastra et al. (2025) – *Computers in Biology and Medicine***
    - **Severe Execution Latency:** 3D CFD and FEA simulations require hours to days on GPU clusters, ruling out real-time point-of-care use.
    - **Invasive Diagnostics:** High-dimensional models demand invasive arterial catheterization or expensive CT scans for baseline parameters.
    - **Vendor Lock-In:** Heavily dependent on proprietary simulation platforms (e.g., ANSYS, MATLAB), creating steep licensing barriers.
- **Vallée (2025) – *JMIR Medical Informatics***
    - Rigid Static Parameters: Pure ODE formulations assume fixed physiological constants, failing to adjust for longitudinal patient drift.
    - Lack of Adaptive ML: Deterministic mathematical equations cannot self-tune using noisy real-time telemetry inputs without manual calibration.

    - Multi-Organ Bottlenecks: Differential equation complexity escalates rapidly, making multi-organ expansion computationally prohibitive.

- **John et al. (2025) – *World Academy of Sciences Journal***
    - Narrow Diagnostic Validation: Practical evaluation is restricted to chronic heart failure, lacking generalizability across acute hypertensive cases.
    - Commercial Software Costs: Relies on commercial simulation tools, restricting open-source modification and EHR interoperability.
    - Static Dosing Constraints: Lacks an interactive "what-if" sandbox for instant, real-time dosage recalibration.
- **Antoun et al. (2026) – *Frontiers in Cardiovascular Medicine***
    - High Sequencing Costs: Integrating multi-omics and Polygenic Risk Scores (PRS) is financially unviable for routine outpatient care.
    - Static Markers: Genomics offer valuable static baseline risk profiles but cannot reflect acute, short-term hemodynamic fluctuations.
    - Ethical Constraints: Aggregating genomic profiles poses strict regulatory compliance challenges regarding genetic discrimination.
- **Obi (2026) – *SSRN Electronic Journal***
    - Theoretical Blueprint Only: Offers an overarching conceptual framework without presenting a validated, functional software engine.
    - Unmeasurable Parameters: Multi-scale modeling requires micro-level biological variables that cannot be extracted during standard patient visits.
    - Unclear Regulatory Path: Highlights SaMD regulatory challenges without presenting concrete strategies for FDA/EMA certification.
- **Kaur et al. (2025) – *IJERAMS***
    - EHR Interoperability Barriers: Legacy hospital information systems cause data format incompatibility and synchronization delays.
    - Heavy Compute Burden: Multimodal foundation models and Graph Neural Networks require high-end local enterprise servers.
    - Domain Mismatch: Designed primarily for oncology kinetics rather than high-speed cardiac hemodynamics.
- **Skalidis et al. (2025) – *Medicina***
    - Sensor Inaccuracy: Consumer-grade cuffless blood pressure wearables suffer from large measurement variance.
    - Adherence Degradation: Long-term patient engagement drops significantly over time, creating gaps in continuous datasets.

- Hypotension Risk: Rule-based titration engines without real-time physical simulation risk over-titration and iatrogenic hypotension.

PROBLEM STATEMENT

Current cardiovascular diagnostic and therapeutic approaches rely heavily on population-averaged clinical guidelines and static diagnostic snapshots. Because these standardized protocols fail to capture inter-individual biological variability—such as differences in vascular compliance, metabolic clearance rates, and genetic profiles—two patients with identical diagnoses often experience vastly different outcomes under the same treatment regimen.

This reliance on generalized clinical data creates significant operational and safety challenges across healthcare systems:

- Reactive Prescribing and Clinical Inertia: Outpatient titration of antihypertensive and inotropic medications depends on retrospective monitoring over weeks or months. This trial-and-error process leads to prolonged periods of suboptimal dosing and high rates of avoidable Adverse Drug Reactions (ADRs).
- Resource-Intensive Clinical Evaluation: Physical clinical trials and treatment evaluations require extensive volunteer recruitment, prolonged execution timelines, and severe financial expenditure, delaying the delivery of novel therapies to patients.
- High Computational Latency of Spatial Models: Existing high-fidelity 3D Computational Fluid Dynamics (CFD) and Finite Element Analysis (FEA) simulations require hours to days of execution time on high-performance computing clusters, rendering real-time bedside utility impossible.
- Demographic and Population Bias: Benchmark clinical prediction tools are predominantly calibrated on homogeneous Western cohorts, leading to reduced predictive accuracy when applied to genetically and lifestyle-diverse populations such as South Asian cohorts.

Therefore, a critical need exists for an accessible, low-latency computational system capable of modeling an individual patient's specific cardiovascular condition and simulating personalized treatment responses *in silico* prior to clinical administration.

PROJECT OBJECTIVES

Primary Goal

The central objective of the DigiTwin project is to design, develop, and validate an interactive, low-cost, and open-source Cardiac Digital Twin and Clinical Trials Simulator. By integrating hybrid reduced-order differential physics solvers with temporal machine learning models, the platform enables safe, non-invasive, cost-effective, and accelerated drug discovery and personalized treatment evaluation.

Key System Objectives

- Develop a Low-Latency Hybrid Physics-AI Engine:
    - Implement a reduced-order Lumped 0D Pharmacokinetic-Pharmacodynamic (PK-PD) differential solver (SciPy ODE, Chi PK-PD) to capture baseline physiological fluid dynamics.
    - Integrate temporal machine learning algorithms (GRU, LSTM, XGBoost) to forecast patient-specific hemodynamic adjustments with a simulation execution time of under 2 seconds on standard web infrastructure.
- Enable Safe and Non-Invasive Virtual Drug Trials:
    - Provide an *in silico* clinical trial environment for pharmaceutical researchers and clinicians to test new drug candidates, dosage curves, and multi-drug interactions without physical risk to human subjects.
- Democratize Precision Cardiology & Reduce Costs:
    - Build the platform entirely on an open-source technical stack (Python, FastAPI, React.js, SQLite) to eliminate reliance on expensive commercial software licenses.
    - Eliminate high financial, infrastructure, and volunteer recruitment barriers associated with early-stage physical clinical trials.
- Address Population Diversity & Demographic Bias:
    - Incorporate localized parameter-seeding algorithms tailored specifically for South Asian / Indian demographic, genetic, and lifestyle baselines, ensuring accurate risk predictions across diverse cohorts.

PROPOSED DESIGN

Step 1: Clinician Input & Presentation Layer

Interface Technologies: Built using React.js and Tailwind CSS for responsive web delivery.

Functionality: Clinicians enter initial Electronic Health Record (EHR) parameters, such as patient age, biological sex, resting Systolic/Diastolic blood pressure, Heart Rate, and Ejection Fraction.

Interactive Control: Interactive sliders allow clinicians to adjust medication choices and real-time dosage curves for virtual evaluation.

Step 2: REST API Middleware & Session Routing

Middleware Engine: Powered by Python FastAPI to manage asynchronous client requests.

Functionality: Receives client HTTP POST payloads, validates patient input parameters, manages active clinical simulation sessions, and routes parsed data to downstream analytical modules.

Step 3: Demographic Seeding Pipeline

Population Calibration: Designed specifically to address Western data bias in traditional models.

Functionality: Machine learning models adjust baseline arterial resistance (R) and vascular compliance (C) based on South Asian / Indian ancestral, genetic, dietary, and lifestyle markers. This step ensures accurate physiological baseline initialization before physics simulation starts.

Step 4: Hybrid Physics-AI Simulation Core

Lumped 0D PK-PD ODE Solver: Computes fundamental hemodynamic fluid mechanics using reduced-order Windkessel differential equations solved via scipy.integrate.solve_ivp and Chi PK-PD engines.

Deep Recurrent AI Predictor: Passes differential outputs through Recurrent Neural Networks (GRU/LSTM) and XGBoost regressors to capture non-linear temporal trends, drug-to-drug interactions, and long-term physiological drift.

Performance: Execution finishes in under 2 seconds, replacing multi-hour 3D CFD simulations.

Step 5: Data Serialization & Session Logging

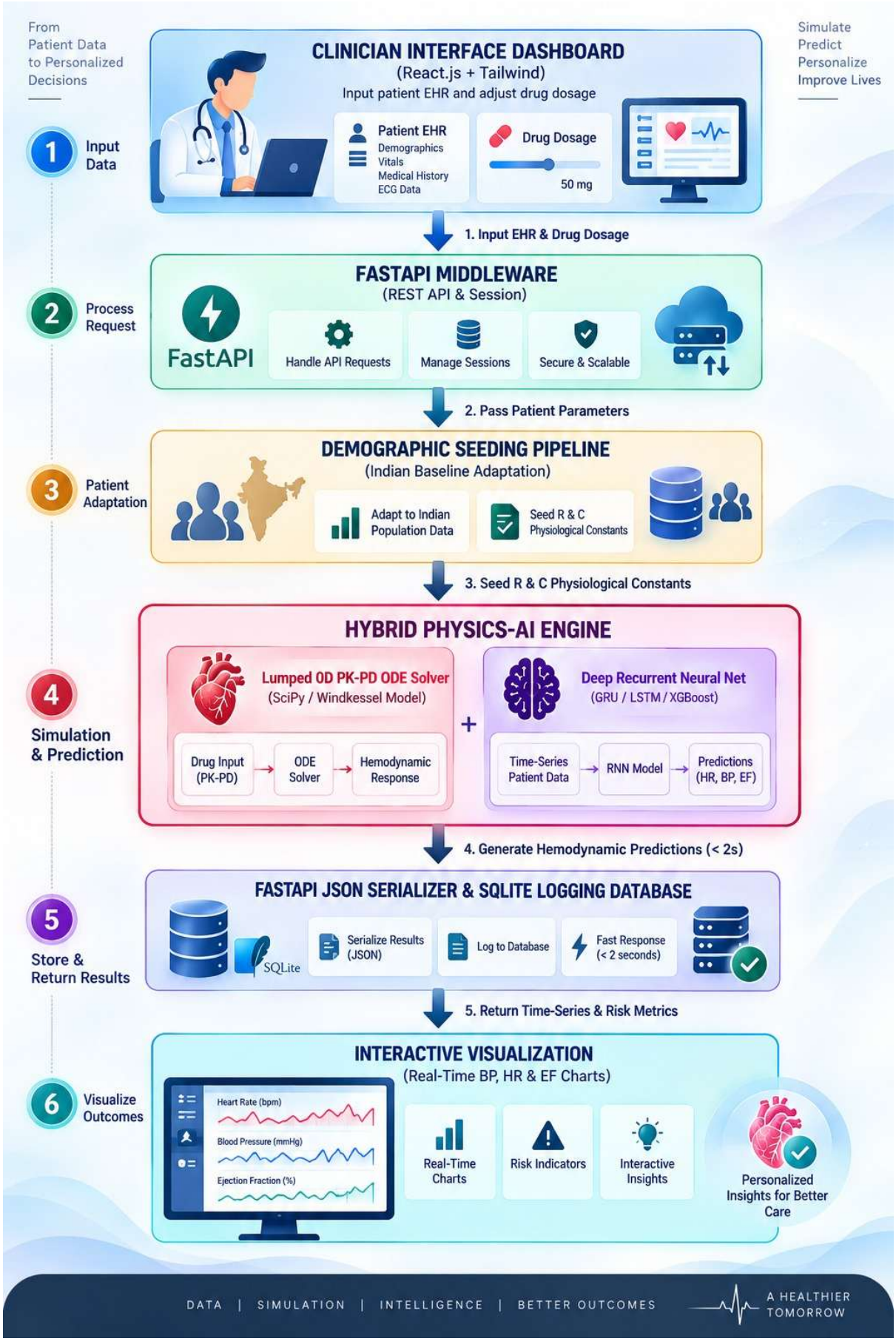
Serialization: Converts raw numerical arrays and model output matrices into lightweight JSON responses via FastAPI.

Persistence: Records simulation inputs, generated risk metrics, and time-series outputs into an SQLite database for clinical auditing and retrospective review.

Step 6: Interactive Dashboard Visualization

Real-Time Display: Transmits the JSON response payload back to the React client.

Output Metrics: Dynamic interactive charts visualize projected Blood Pressure (BP) curves, Heart Rate (HR) variations, Ejection Fraction (EF) adjustments, and Adverse Drug Reaction (ADR) risk indicators.


From Patient Data to Personalized Decisions
Simulate Predict Personalize Improve Lives
1 Input Data
CLINICIAN INTERFACE DASHBOARD
(React.js + Tailwind)
Input patient EHR and adjust drug dosage
Patient EHR
Demographics
Vitals
Medical History
ECG Data
Drug Dosage
50 mg
1. Input EHR & Drug Dosage
2 Process Request
FASTAPI MIDDLEWARE
(REST API & Session)
FastAPI
Handle API Requests
Manage Sessions
Secure & Scalable
2. Pass Patient Parameters
3 Patient Adaptation
DEMOGRAPHIC SEEDING PIPELINE
(Indian Baseline Adaptation)
Adapt to Indian Population Data
Seed R & C Physiological Constants
3. Seed R & C Physiological Constants
4 Simulation & Prediction
HYBRID PHYSICS-AI ENGINE
Lumped 0D PK-PD ODE Solver
(SciPy / Windkessel Model)
Drug Input (PK-PD)
ODE Solver
Hemodynamic Response
+
Deep Recurrent Neural Net
(GRU / LSTM / XGBoost)
Time-Series Patient Data
RNN Model
Predictions (HR, BP, EF)
4. Generate Hemodynamic Predictions (< 2s)
5 Store & Return Results
FASTAPI JSON SERIALIZER & SQLITE LOGGING DATABASE
SQLite
Serialize Results (JSON)
Log to Database
Fast Response (< 2 seconds)
5. Return Time-Series & Risk Metrics
6 Visualize Outcomes
INTERACTIVE VISUALIZATION
(Real-Time BP, HR & EF Charts)
Heart Rate (bpm)
Blood Pressure (mmHg)
Ejection Fraction (%)
Real-Time Charts
Risk Indicators
Interactive Insights
Personalized Insights for Better Care
DATA | SIMULATION | INTELLIGENCE | BETTER OUTCOMES
A HEALTHIER TOMORROW

EXPECTED OUTCOMES

The **DigiTwin** platform is designed to achieve measurable, high-impact outcomes across computational performance, clinical workflows, economic sustainability, and demographic equity:

Sub-2-Second Computational Latency

Replaces computationally heavy 3D spatial simulations with a reduced-order 0D Lumped PK-PD differential solver and machine learning surrogates. This reduces execution times from hours or days on high-performance computing clusters to **under 2 seconds** on standard web infrastructure, enabling interactive, real-time bedside decision support.

30–40% Reduction in Adverse Drug Reactions (ADRs)

Provides clinicians with a pre-prescriptive *"Simulate before you prescribe"* virtual sandbox. By forecasting individual therapeutic responses, drug-to-drug interactions, and toxicity thresholds *in silico*, the platform minimizes trial-and-error prescribing and prevents severe treatment-induced complications.

Demographically Inclusive Predictive Modeling

Overcomes the population biases inherent in standard Western-calibrated clinical tools (such as MIMIC-IV) by implementing a localized parameter-seeding engine. This pipeline dynamically adjusts baseline vascular resistance (R) and compliance (C) based on South Asian / Indian ancestral, dietary, genetic, and lifestyle factors.

Zero-License-Cost Open-Source Stack

Constructed entirely using open-source technologies (FastAPI, React.js, SciPy, TensorFlow/PyTorch, SQLite). Eliminating high-cost proprietary software dependencies (e.g., ANSYS, MATLAB) enables seamless, low-cost deployment across resource-constrained public hospitals and rural healthcare settings.

Optimized Clinical Workflows and Cost Reduction

Automating drug response projections saves healthcare providers an estimated **2 hours per week** in manual dose titration management. Furthermore, early risk identification and prevented emergency hospital readmissions are projected to save **~₹7 lakh annually per high-risk cardiovascular patient**.

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